A. Let $F : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $F(x_1, x_2, x_3) = (\frac{x_1}{1+x_3}, \frac{x_2}{1+x_3})$. Let x by defined by $x(u, v) = (\frac{2u}{u^2+v^2+1}, \frac{2v}{u^2+v^2+1}, \frac{-(u^2+v^2-1)}{u^2+v^2+1})$. We compute that $F(x(u, v)) = (u, v)$.

B. .A *x is regular when α is regular.*

Proof. We compute $\frac{\partial x}{\partial u} = (\frac{\partial \alpha_1}{\partial u}, \frac{\partial \alpha_2}{\partial u}, 0)$ and $\frac{\partial x}{\partial v} = (0, 0, 1)$. For $\frac{\partial x}{\partial u}$ and $\frac{\partial x}{\partial v}$ to be linearly independent on all of the surface, we see that $(\alpha_1^{'}, \alpha_2^{'}) \neq (0, 0)$. So, it must be the case that α is regular. □

.B The image of x is a surface when α behaves nicely. For example, if α self-intersects, the image of x is not a surface.

C. .A Note that in general, $f(x) = x^T M x = M_{11}{x_1}^2 + M_{22}{x_2}^2 + M_{33}{x_3}^2 + 2M_{12}x_1x_2 + 2M_{13}x_1x_3 + 2M_{23}x_2x_3$. When $M = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $f(x) = x_3^2 = 1$. This gives us an x_1x_2-plane at $x_3 = \pm 1$. When $M = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $f(x) = 1 = x_2^2 + x_3^2$. This creates a unbounded cylinder along the x_1 axis. When $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $f(x) = 1 = x_1^2 + x_2^2 + x_3^2$. This creates the unit sphere.

.B *$S = \{x | f(x) = x^T M x = 1\}$ is non-empty when M has at least one positive eigenvalue.*

Proof. Let λ be a positive eigenvalue of M, i.e. there is eigenvector v that satisfies $Mv = \lambda v$. We can then consider $f(cv)$ with $c \in \mathbb{R}$. We compute

$$f(cv) = (cv)^T M(cv) = cv^T c\lambda v = c^2 |v|^2 \lambda.$$

We want $c^2 |v|^2 \lambda = 1$, so solving, we get that $c = \frac{1}{\sqrt{|v|^2 \lambda}}$. We know this c exists because λ is positive. So, when $x = \frac{v}{\sqrt{|v|^2 \lambda}}$ where λ, v are a positive eigenvalue of M and its corresponding eigenvector, $f(x) = 1$, and therefore S is non-empty. □